Examiner only

8. Hydrogen peroxide solution, H_2O_2, is used in commercial stain removers.

A GCSE class investigated how effective four stain removers are at removing stains. Stain removers **A**, **B**, **C** and **D** contain different concentrations of hydrogen peroxide.

The students tested how effective each one is at removing an identical oil stain from four towels.

Their findings are outlined below.

Stain remover **A**	
Working temperature	= 50 °C
Cost per 100 cm^3	= 99p
Time to remove stain	= 40 min
Volume needed	= 20 cm^3

Stain remover **B**	
Working temperature	= 30 °C
Cost per 100 cm^3	= £1.99
Time to remove stain	= 40 min
Volume needed	= 10 cm^3

Stain remover **C**	
Working temperature	= 20 °C
Cost per 100 cm^3	= £2.49
Time to remove stain	= 20 min
Volume needed	= 5 cm^3

Stain remover **D**	
Working temperature	= 30 °C
Cost per 100 cm^3	= £1.49
Time to remove stain	= 30 min
Volume needed	= 10 cm^3

(a) In carrying out this investigation, which variables were kept the same in order to get valid results? Tick (✓) the correct answer. [1]

type of oil used, towel material and volume of hydrogen peroxide ☐

type of oil used, towel material and temperature of stain remover ☐

type of oil used and towel material ☐

type of oil used, towel material and cost of stain remover ☐

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(b) Tick (✓) **all** of the statements which **could** explain why stain remover **A** has to be heated to 50 °C before it removes the stain. [1]

it is the cheapest stain remover	☐
it is heat resistant	☐
it has a low concentration of hydrogen peroxide	☐
it takes a long time to work	☐

(c) The students found that stain removers **B** and **D** used the same volume and worked best at the same temperature.

Assuming that they have the same hydrogen peroxide concentration, suggest a possible reason why **D** removes the stain more quickly than **B**. [1]

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(d) One student went on to investigate the decomposition of hydrogen peroxide.

The equation for the reaction is as follows.

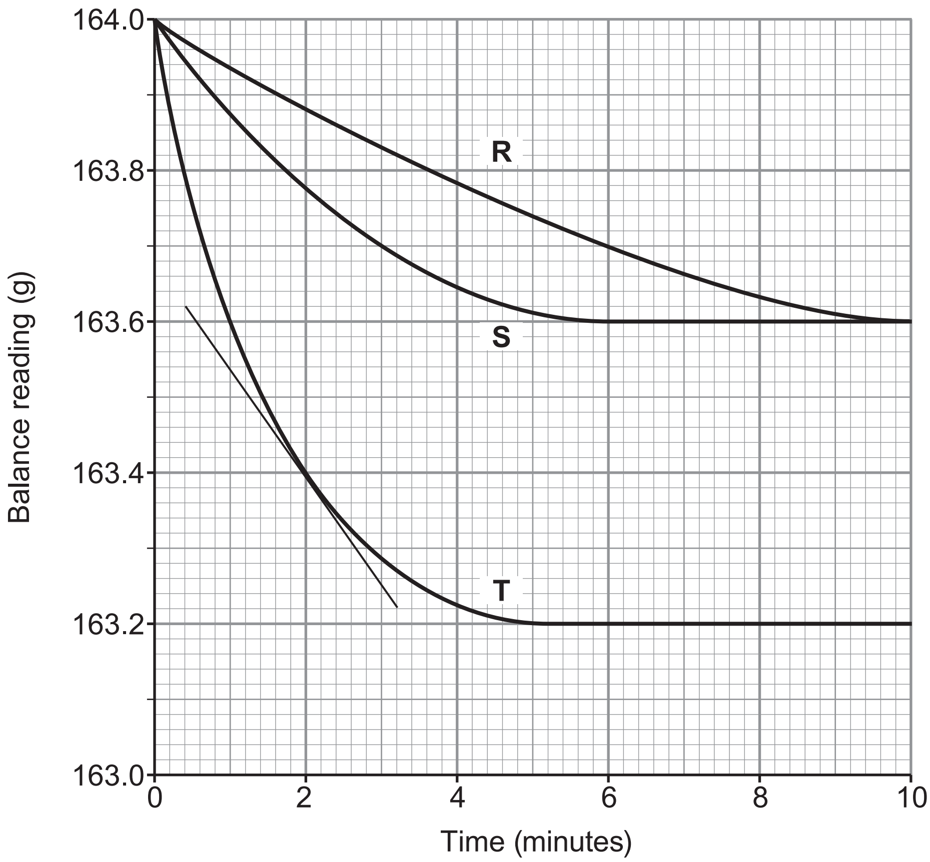
$$2H_2O_2 \longrightarrow 2H_2O + O_2$$

The student investigated the effect of changing the concentration of the hydrogen peroxide solution on the rate of the reaction. She used manganese dioxide as a catalyst in each experiment.

This is the method she used.

- Pour 50 cm^3 of hydrogen peroxide solution of concentration **R** into a conical flask on a digital balance.
- Add 1 g of catalyst and place some cotton wool loosely in the neck of the flask. Record the balance reading and immediately start a stopwatch.
- Record the balance reading every minute until the mass no longer changes.
- Carry out the experiment twice more using hydrogen peroxide of different concentrations, **S** and **T**.

Her results are plotted on the grid below.



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(i) Using the tangent shown on the graph, calculate the rate of reaction for concentration **T** at 2 minutes. Show your working. [2]

Rate at 2 minutes = ... g/minute

(ii) The initial rate for concentration **S** is half the initial rate for concentration **T**. Explain this difference in rate in terms of the particle theory. [3]

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END OF PAPER